

Supplementary Information for:
Predicted Gene-Program Activations from Histology Are Fragile to
Routine Image Perturbations, and the Failure Mode Depends on
the Encoder

Abdelrahman A. Alshames

This file contains supplementary tables referenced in the main text. All numbers here are produced by the same pipeline and scripts described in the main text’s Methods section and cited from the same repository (see Code Availability).

Table S4: CONCH: linear-offset correction’s relative GPFR reduction (main text Section 2.7), summarized per perturbation across its 5 programs. Positive = correction lowered GPFR; negative = made it worse. `jpeg`’s min (-133%) is the largest actual worsening observed – the correction is not a reliably-signed effect even within one perturbation, let alone across the suite.

Perturbation	Mean reduction	Min	Max
brightness	34%	14%	62%
contrast	37%	12%	53%
gamma	23%	-33%	68%
stain_shift	8%	-22%	43%
blur	7%	-29%	47%
jpeg	-14%	-133%	56%
scanner_color	31%	5%	65%

Table S1: CONCH: Gene-Program Flip Rate (GPFR) per perturbation/program on IDC, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
scanner_color	hypoxia	0.390	[0.329, 0.575]	0.0006*
scanner_color	immune_activation	0.201	[0.050, 0.524]	0.0006*
jpeg	hypoxia	0.190	[0.108, 0.393]	0.0006*
brightness	hypoxia	0.183	[0.122, 0.302]	0.0006*
stain_shift	hypoxia	0.160	[0.078, 0.308]	0.0006*
contrast	hypoxia	0.158	[0.076, 0.275]	0.0006*
scanner_color	angiogenesis	0.157	[0.065, 0.277]	0.0006*
jpeg	stromal_ecm	0.128	[0.087, 0.205]	0.0006*
blur	hypoxia	0.121	[0.077, 0.200]	0.0006*
blur	angiogenesis	0.117	[0.082, 0.214]	0.0006*
contrast	angiogenesis	0.112	[0.049, 0.185]	0.0006*
stain_shift	proliferation	0.094	[0.073, 0.148]	0.0006*
scanner_color	stromal_ecm	0.094	[0.073, 0.118]	0.0006*
scanner_color	proliferation	0.090	[0.054, 0.141]	0.0006*
gamma	angiogenesis	0.083	[0.043, 0.141]	0.0006*
blur	proliferation	0.082	[0.053, 0.136]	0.0006*
jpeg	proliferation	0.076	[0.054, 0.107]	0.0006*
brightness	angiogenesis	0.074	[0.025, 0.134]	0.0006*
jpeg	angiogenesis	0.073	[0.036, 0.120]	0.0006*
stain_shift	angiogenesis	0.058	[0.021, 0.119]	0.0006*
gamma	hypoxia	0.057	[0.020, 0.133]	0.0006*
brightness	proliferation	0.052	[0.028, 0.108]	0.0006*
stain_shift	stromal_ecm	0.042	[0.021, 0.063]	0.0006*
brightness	immune_activation	0.038	[0.004, 0.158]	0.0077*
blur	stromal_ecm	0.038	[0.010, 0.063]	0.0006*
brightness	stromal_ecm	0.032	[0.015, 0.062]	0.0006*
contrast	proliferation	0.026	[0.007, 0.056]	0.0028*
contrast	stromal_ecm	0.024	[0.020, 0.030]	0.0006*
gamma	proliferation	0.022	[0.006, 0.049]	0.0044*
contrast	immune_activation	0.018	[0.013, 0.057]	0.0006*
jpeg	immune_activation	0.012	[0.008, 0.078]	0.0006*
gamma	stromal_ecm	0.011	[0.005, 0.028]	0.0006*
blur	immune_activation	0.010	[0.004, 0.055]	0.0053*
stain_shift	immune_activation	0.008	[0.002, 0.022]	0.0006*
gamma	immune_activation	0.003	[0.000, 0.007]	0.0655

Table S2: UNI: Gene-Program Flip Rate (GPFR) per perturbation/program on IDC, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
jpeg	immune_activation	0.262	[0.108, 0.638]	0.0006*
scanner_color	hypoxia	0.198	[0.139, 0.386]	0.0006*
scanner_color	immune_activation	0.169	[0.011, 0.517]	0.0006*
jpeg	hypoxia	0.151	[0.121, 0.217]	0.0006*
jpeg	proliferation	0.128	[0.050, 0.293]	0.0006*
stain_shift	proliferation	0.106	[0.030, 0.233]	0.0006*
stain_shift	hypoxia	0.090	[0.047, 0.178]	0.0006*
gamma	hypoxia	0.079	[0.036, 0.175]	0.0006*
contrast	angiogenesis	0.079	[0.038, 0.126]	0.0006*
jpeg	angiogenesis	0.076	[0.064, 0.093]	0.0006*
gamma	angiogenesis	0.072	[0.018, 0.186]	0.0006*
stain_shift	angiogenesis	0.066	[0.024, 0.109]	0.0006*
scanner_color	proliferation	0.056	[0.030, 0.147]	0.0006*
scanner_color	angiogenesis	0.053	[0.038, 0.082]	0.0006*
brightness	angiogenesis	0.051	[0.035, 0.077]	0.0006*
scanner_color	stromal_ecm	0.049	[0.029, 0.070]	0.0006*
stain_shift	stromal_ecm	0.046	[0.016, 0.078]	0.0006*
jpeg	stromal_ecm	0.038	[0.026, 0.059]	0.0006*
brightness	hypoxia	0.036	[0.021, 0.074]	0.0006*
gamma	immune_activation	0.034	[0.017, 0.155]	0.0006*
contrast	hypoxia	0.032	[0.009, 0.096]	0.0006*
gamma	stromal_ecm	0.029	[0.016, 0.044]	0.0006*
contrast	immune_activation	0.024	[0.013, 0.114]	0.0006*
contrast	stromal_ecm	0.024	[0.009, 0.046]	0.0006*
blur	proliferation	0.020	[0.003, 0.048]	0.0006*
brightness	proliferation	0.018	[0.013, 0.032]	0.0006*
blur	hypoxia	0.014	[0.005, 0.049]	0.0006*
brightness	stromal_ecm	0.013	[0.009, 0.024]	0.0006*
gamma	proliferation	0.013	[0.002, 0.053]	0.0048*
contrast	proliferation	0.011	[0.002, 0.044]	0.0070*
blur	angiogenesis	0.011	[0.000, 0.023]	0.0459
stain_shift	immune_activation	0.006	[0.000, 0.026]	0.0631
brightness	immune_activation	0.004	[0.000, 0.046]	0.0412
blur	stromal_ecm	0.001	[0.000, 0.002]	0.4548
blur	immune_activation	0.000	[0.000, 0.000]	1.0000

Table S3: Spearman rank correlation of each z_{thresh} ’s GPFR ranking (across all 35 perturbation×program combinations) against the default $z_{\text{thresh}} = 1.0$ ranking, for both encoders on IDC.

z_{thresh}	CONCH ρ	UNI ρ
0.5	0.965	0.942
0.75	0.989	0.975
1.0	1.000	1.000
1.5	0.946	0.916
2.0	0.858	0.700

Table S5: Hallmark gene-set coverage on IDC’s 280-gene common panel – the intersection across all 4 IDC samples’ individual panels (288/321/280/280 genes for NCBI783/NCBI785/TENX95/TENX99 respectively; not identical across samples), i.e. exactly the gene set the modeling pipeline scores against. “Shared” counts genes that also belong to at least one other of the 5 scored programs. At this coverage, **proliferation** and **angiogenesis** are effectively small marker panels rather than comprehensive pathway scores.

Program	Genes present / total	Coverage	Shared with other programs
immune_activation	16/363	4.4%	1/16
proliferation	6/327	1.8%	0/6
hypoxia	8/200	4.0%	4/8
stromal_ecm	19/200	9.5%	3/19
angiogenesis	5/36	13.9%	4/5

Table S6: Stain-shift magnitude sweep on IDC (CONCH), including the **stain_shift_identity** null control (a full Macenko decompose/reconstruct round trip with NO intended stain-vector change). Any nonzero GPFR at Identity is reconstruction noise, not a real color-shift effect; the gap between Identity and Moderate is the actual effect size.

Magnitude	Mean SSIM	Mean GPFR	Max GPFR
Identity (null control)	0.981	0.040	0.108
Mild (h=1.10/e=0.90)	0.978	0.043	0.106
Moderate (h=1.30/e=0.70)	0.954	0.072	0.160

Table S7: CONCH: Gene-Program Flip Rate (GPFR) per perturbation/program on COAD, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
scanner_color	hypoxia	0.620	[0.559, 0.724]	0.0006*
jpeg	hypoxia	0.301	[0.249, 0.396]	0.0006*
blur	immune_activation	0.284	[0.207, 0.373]	0.0006*
brightness	hypoxia	0.221	[0.174, 0.286]	0.0006*
brightness	proliferation	0.204	[0.088, 0.379]	0.0006*
scanner_color	proliferation	0.193	[0.150, 0.280]	0.0006*
brightness	stromal_ecm	0.168	[0.142, 0.235]	0.0006*
blur	hypoxia	0.163	[0.144, 0.223]	0.0006*
scanner_color	angiogenesis	0.157	[0.100, 0.245]	0.0006*
blur	angiogenesis	0.149	[0.057, 0.257]	0.0006*
scanner_color	stromal_ecm	0.148	[0.064, 0.279]	0.0006*
scanner_color	immune_activation	0.138	[0.076, 0.228]	0.0006*
contrast	hypoxia	0.128	[0.066, 0.222]	0.0006*
brightness	angiogenesis	0.116	[0.092, 0.153]	0.0006*
blur	stromal_ecm	0.116	[0.055, 0.173]	0.0006*
jpeg	immune_activation	0.112	[0.087, 0.137]	0.0006*
stain_shift	hypoxia	0.112	[0.069, 0.184]	0.0006*
contrast	proliferation	0.094	[0.030, 0.215]	0.0067*
stain_shift	angiogenesis	0.082	[0.045, 0.117]	0.0006*
jpeg	angiogenesis	0.080	[0.048, 0.125]	0.0006*
brightness	immune_activation	0.078	[0.027, 0.142]	0.0006*
jpeg	stromal_ecm	0.072	[0.051, 0.137]	0.0006*
contrast	angiogenesis	0.069	[0.036, 0.108]	0.0006*
blur	proliferation	0.069	[0.058, 0.136]	0.0006*
gamma	hypoxia	0.069	[0.049, 0.119]	0.0006*
gamma	angiogenesis	0.064	[0.033, 0.109]	0.0006*
jpeg	proliferation	0.052	[0.033, 0.122]	0.0006*
contrast	stromal_ecm	0.048	[0.007, 0.092]	0.0006*
contrast	immune_activation	0.044	[0.004, 0.104]	0.0064*
stain_shift	proliferation	0.039	[0.028, 0.058]	0.0006*
stain_shift	stromal_ecm	0.034	[0.015, 0.044]	0.0006*
stain_shift	immune_activation	0.032	[0.024, 0.044]	0.0006*
gamma	immune_activation	0.012	[0.002, 0.026]	0.0033*
gamma	stromal_ecm	0.012	[0.007, 0.023]	0.0006*
gamma	proliferation	0.001	[0.000, 0.013]	0.1384

Table S8: UNI: Gene-Program Flip Rate (GPFR) per perturbation/program on COAD, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
scanner_color	angiogenesis	0.213	[0.175, 0.256]	0.0006*
stain_shift	hypoxia	0.202	[0.156, 0.293]	0.0006*
jpeg	hypoxia	0.193	[0.136, 0.293]	0.0006*
scanner_color	hypoxia	0.191	[0.152, 0.281]	0.0006*
scanner_color	immune_activation	0.158	[0.101, 0.239]	0.0006*
jpeg	angiogenesis	0.154	[0.137, 0.171]	0.0006*
jpeg	stromal_ecm	0.153	[0.099, 0.247]	0.0006*
jpeg	immune_activation	0.137	[0.094, 0.182]	0.0006*
brightness	hypoxia	0.110	[0.101, 0.141]	0.0006*
scanner_color	proliferation	0.094	[0.049, 0.236]	0.0006*
stain_shift	angiogenesis	0.089	[0.049, 0.114]	0.0006*
stain_shift	proliferation	0.089	[0.052, 0.227]	0.0006*
scanner_color	stromal_ecm	0.083	[0.048, 0.131]	0.0006*
contrast	hypoxia	0.077	[0.048, 0.147]	0.0006*
gamma	angiogenesis	0.073	[0.031, 0.109]	0.0006*
stain_shift	stromal_ecm	0.064	[0.025, 0.202]	0.0006*
gamma	hypoxia	0.063	[0.027, 0.133]	0.0006*
stain_shift	immune_activation	0.053	[0.030, 0.079]	0.0006*
brightness	angiogenesis	0.050	[0.029, 0.062]	0.0006*
blur	hypoxia	0.047	[0.043, 0.090]	0.0006*
jpeg	proliferation	0.036	[0.029, 0.096]	0.0006*
gamma	stromal_ecm	0.036	[0.010, 0.063]	0.0006*
contrast	angiogenesis	0.029	[0.014, 0.048]	0.0006*
gamma	immune_activation	0.024	[0.002, 0.049]	0.0006*
blur	angiogenesis	0.021	[0.012, 0.031]	0.0006*
gamma	proliferation	0.019	[0.007, 0.065]	0.0006*
brightness	proliferation	0.012	[0.004, 0.051]	0.0006*
contrast	stromal_ecm	0.010	[0.003, 0.017]	0.0087*
contrast	immune_activation	0.010	[0.003, 0.018]	0.0006*
brightness	immune_activation	0.009	[0.007, 0.015]	0.0006*
brightness	stromal_ecm	0.006	[0.000, 0.030]	0.0694
blur	stromal_ecm	0.004	[0.000, 0.022]	0.0694
contrast	proliferation	0.003	[0.000, 0.016]	0.0497
blur	immune_activation	0.002	[0.000, 0.004]	0.1934
blur	proliferation	0.000	[0.000, 0.008]	0.6257

Table S9: CONCH: Gene-Program Flip Rate (GPFR) per perturbation/program on IDC, leave-one-slide-out, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
scanner_color	immune_activation	0.184	[0.035, 0.419]	0.0005*
scanner_color	hypoxia	0.175	[0.116, 0.328]	0.0005*
jpeg	stromal_ecm	0.123	[0.066, 0.213]	0.0005*
scanner_color	angiogenesis	0.113	[0.083, 0.261]	0.0005*
scanner_color	stromal_ecm	0.111	[0.067, 0.199]	0.0005*
blur	angiogenesis	0.093	[0.082, 0.162]	0.0005*
blur	hypoxia	0.091	[0.075, 0.161]	0.0005*
jpeg	proliferation	0.087	[0.040, 0.160]	0.0005*
contrast	angiogenesis	0.085	[0.060, 0.122]	0.0005*
scanner_color	proliferation	0.085	[0.044, 0.166]	0.0005*
jpeg	hypoxia	0.082	[0.049, 0.175]	0.0005*
stain_shift	stromal_ecm	0.082	[0.051, 0.113]	0.0005*
brightness	stromal_ecm	0.078	[0.026, 0.167]	0.0005*
stain_shift	proliferation	0.077	[0.016, 0.165]	0.0005*
stain_shift	angiogenesis	0.074	[0.041, 0.101]	0.0005*
gamma	angiogenesis	0.074	[0.042, 0.104]	0.0005*
jpeg	angiogenesis	0.070	[0.066, 0.086]	0.0005*
contrast	hypoxia	0.067	[0.037, 0.160]	0.0005*
brightness	hypoxia	0.065	[0.022, 0.169]	0.0005*
brightness	angiogenesis	0.061	[0.033, 0.118]	0.0005*
blur	proliferation	0.061	[0.038, 0.101]	0.0005*
stain_shift	hypoxia	0.053	[0.033, 0.091]	0.0005*
blur	stromal_ecm	0.049	[0.030, 0.071]	0.0005*
brightness	immune_activation	0.043	[0.005, 0.139]	0.0005*
brightness	proliferation	0.034	[0.018, 0.074]	0.0005*
contrast	stromal_ecm	0.031	[0.014, 0.070]	0.0005*
contrast	proliferation	0.027	[0.013, 0.064]	0.0005*
jpeg	immune_activation	0.027	[0.016, 0.059]	0.0005*
gamma	hypoxia	0.027	[0.015, 0.062]	0.0005*
blur	immune_activation	0.026	[0.009, 0.050]	0.0005*
gamma	proliferation	0.026	[0.009, 0.073]	0.0005*
gamma	stromal_ecm	0.019	[0.005, 0.040]	0.0005*
stain_shift	immune_activation	0.017	[0.016, 0.028]	0.0005*
contrast	immune_activation	0.017	[0.007, 0.051]	0.0005*
gamma	immune_activation	0.005	[0.000, 0.022]	0.0580

Table S10: UNI: Gene-Program Flip Rate (GPFR) per perturbation/program on IDC, leave-one-slide-out, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
jpeg	immune_activation	0.317	[0.106, 0.630]	0.0005*
scanner_color	immune_activation	0.261	[0.006, 0.564]	0.0005*
jpeg	proliferation	0.194	[0.012, 0.440]	0.0005*
stain_shift	stromal_ecm	0.155	[0.048, 0.298]	0.0005*
jpeg	hypoxia	0.152	[0.135, 0.346]	0.0005*
scanner_color	stromal_ecm	0.134	[0.070, 0.250]	0.0005*
scanner_color	hypoxia	0.132	[0.047, 0.397]	0.0005*
jpeg	angiogenesis	0.125	[0.085, 0.191]	0.0005*
scanner_color	angiogenesis	0.122	[0.097, 0.153]	0.0005*
contrast	angiogenesis	0.108	[0.070, 0.140]	0.0005*
stain_shift	proliferation	0.092	[0.016, 0.231]	0.0005*
stain_shift	hypoxia	0.083	[0.034, 0.255]	0.0005*
brightness	angiogenesis	0.083	[0.071, 0.111]	0.0005*
scanner_color	proliferation	0.073	[0.026, 0.230]	0.0005*
stain_shift	angiogenesis	0.065	[0.033, 0.112]	0.0005*
gamma	immune_activation	0.060	[0.014, 0.152]	0.0005*
jpeg	stromal_ecm	0.053	[0.028, 0.076]	0.0005*
gamma	angiogenesis	0.052	[0.046, 0.087]	0.0005*
gamma	hypoxia	0.037	[0.032, 0.121]	0.0005*
gamma	proliferation	0.035	[0.005, 0.117]	0.0005*
gamma	stromal_ecm	0.029	[0.011, 0.041]	0.0005*
blur	angiogenesis	0.027	[0.007, 0.056]	0.0005*
contrast	immune_activation	0.027	[0.010, 0.075]	0.0005*
brightness	hypoxia	0.025	[0.021, 0.056]	0.0005*
contrast	stromal_ecm	0.024	[0.009, 0.043]	0.0005*
contrast	hypoxia	0.023	[0.015, 0.082]	0.0005*
blur	proliferation	0.019	[0.001, 0.073]	0.0005*
brightness	proliferation	0.016	[0.008, 0.036]	0.0005*
contrast	proliferation	0.015	[0.003, 0.054]	0.0005*
brightness	stromal_ecm	0.010	[0.005, 0.020]	0.0005*
brightness	immune_activation	0.009	[0.004, 0.047]	0.0053*
stain_shift	immune_activation	0.009	[0.001, 0.055]	0.0005*
blur	hypoxia	0.003	[0.002, 0.026]	0.0005*
blur	stromal_ecm	0.001	[0.000, 0.003]	0.2747
blur	immune_activation	0.000	[0.000, 0.001]	0.9290

Table S11: CONCH: Gene-Program Flip Rate (GPFR) per perturbation/program on COAD, leave-one-slide-out, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
scanner_color	hypoxia	0.462	[0.317, 0.690]	0.0005*
scanner_color	proliferation	0.252	[0.195, 0.349]	0.0005*
blur	immune_activation	0.243	[0.177, 0.345]	0.0005*
scanner_color	immune_activation	0.193	[0.129, 0.283]	0.0005*
brightness	proliferation	0.186	[0.094, 0.293]	0.0005*
blur	angiogenesis	0.185	[0.077, 0.466]	0.0005*
blur	hypoxia	0.177	[0.150, 0.257]	0.0005*
jpeg	hypoxia	0.150	[0.118, 0.207]	0.0005*
brightness	hypoxia	0.141	[0.113, 0.202]	0.0005*
brightness	stromal_ecm	0.127	[0.054, 0.206]	0.0005*
stain_shift	hypoxia	0.116	[0.047, 0.199]	0.0005*
blur	proliferation	0.116	[0.061, 0.188]	0.0005*
brightness	immune_activation	0.111	[0.051, 0.214]	0.0005*
blur	stromal_ecm	0.099	[0.061, 0.230]	0.0005*
contrast	hypoxia	0.094	[0.069, 0.153]	0.0005*
scanner_color	angiogenesis	0.091	[0.068, 0.127]	0.0005*
brightness	angiogenesis	0.088	[0.061, 0.125]	0.0005*
scanner_color	stromal_ecm	0.080	[0.046, 0.181]	0.0005*
contrast	proliferation	0.077	[0.032, 0.127]	0.0005*
jpeg	angiogenesis	0.070	[0.056, 0.084]	0.0005*
stain_shift	angiogenesis	0.069	[0.043, 0.097]	0.0005*
jpeg	immune_activation	0.067	[0.041, 0.100]	0.0005*
jpeg	proliferation	0.060	[0.031, 0.118]	0.0005*
contrast	angiogenesis	0.054	[0.032, 0.112]	0.0005*
stain_shift	immune_activation	0.053	[0.012, 0.110]	0.0005*
contrast	immune_activation	0.051	[0.009, 0.130]	0.0005*
gamma	angiogenesis	0.047	[0.032, 0.075]	0.0005*
stain_shift	proliferation	0.041	[0.029, 0.063]	0.0005*
gamma	hypoxia	0.035	[0.032, 0.053]	0.0005*
contrast	stromal_ecm	0.024	[0.005, 0.103]	0.0005*
stain_shift	stromal_ecm	0.020	[0.013, 0.033]	0.0005*
jpeg	stromal_ecm	0.019	[0.012, 0.047]	0.0005*
gamma	immune_activation	0.013	[0.001, 0.037]	0.0030*
gamma	stromal_ecm	0.009	[0.005, 0.024]	0.0005*
gamma	proliferation	0.006	[0.003, 0.013]	0.0005*

Table S12: UNI: Gene-Program Flip Rate (GPFR) per perturbation/program on COAD, leave-one-slide-out, with 95% CI from the 4-slide cluster bootstrap (2000 resamples) and Benjamini-Hochberg FDR-corrected q -value across all 35 tests. * FDR<0.05 AND the CI's lower bound does not round to 0.000 – with only 4 slides to resample, the one-sided bootstrap p -value is quantized (at most $4^4 = 256$ distinct resample outcomes, giving as few as 6–9 distinct p -values across all 35 tests here), so it can reject the boundary null for an interval that still visibly touches zero; requiring both keeps the star consistent with the printed interval.

Perturbation	Program	GPFR	95% CI	FDR q
stain_shift	proliferation	0.165	[0.053, 0.311]	0.0006*
scanner_color	hypoxia	0.159	[0.148, 0.282]	0.0006*
jpeg	hypoxia	0.153	[0.145, 0.337]	0.0006*
scanner_color	immune_activation	0.124	[0.064, 0.189]	0.0006*
scanner_color	proliferation	0.122	[0.037, 0.232]	0.0006*
stain_shift	hypoxia	0.108	[0.055, 0.213]	0.0006*
jpeg	immune_activation	0.105	[0.093, 0.141]	0.0006*
scanner_color	angiogenesis	0.103	[0.065, 0.183]	0.0006*
jpeg	angiogenesis	0.084	[0.050, 0.144]	0.0006*
jpeg	stromal_ecm	0.079	[0.055, 0.113]	0.0006*
contrast	hypoxia	0.075	[0.032, 0.175]	0.0006*
gamma	hypoxia	0.072	[0.029, 0.173]	0.0006*
brightness	hypoxia	0.061	[0.047, 0.105]	0.0006*
scanner_color	stromal_ecm	0.059	[0.036, 0.152]	0.0006*
stain_shift	angiogenesis	0.056	[0.030, 0.087]	0.0006*
jpeg	proliferation	0.051	[0.025, 0.083]	0.0006*
gamma	angiogenesis	0.050	[0.013, 0.084]	0.0006*
stain_shift	stromal_ecm	0.049	[0.019, 0.126]	0.0006*
brightness	angiogenesis	0.044	[0.022, 0.076]	0.0006*
stain_shift	immune_activation	0.033	[0.022, 0.070]	0.0006*
brightness	proliferation	0.026	[0.017, 0.039]	0.0006*
contrast	proliferation	0.013	[0.005, 0.027]	0.0006*
brightness	immune_activation	0.012	[0.007, 0.038]	0.0006*
contrast	angiogenesis	0.011	[0.007, 0.017]	0.0006*
gamma	proliferation	0.010	[0.003, 0.045]	0.0006*
blur	hypoxia	0.010	[0.006, 0.045]	0.0006*
blur	angiogenesis	0.007	[0.002, 0.010]	0.0052*
brightness	stromal_ecm	0.007	[0.004, 0.021]	0.0006*
contrast	stromal_ecm	0.005	[0.000, 0.014]	0.0519
contrast	immune_activation	0.005	[0.001, 0.015]	0.0025*
gamma	immune_activation	0.004	[0.000, 0.010]	0.0203
gamma	stromal_ecm	0.003	[0.001, 0.013]	0.0048*
blur	stromal_ecm	0.001	[0.000, 0.009]	0.2989
blur	proliferation	0.000	[0.000, 0.002]	0.4445
blur	immune_activation	0.000	[0.000, 0.002]	0.8461